Projects, Interview Preparation & Career

Project ideas from beginner to expert, interview question bank and a Kaggle strategy.

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1. Beginner projects (Weeks 1-8) [Beginner]

Explanation (English)
1) Student marks analysis (pandas). 2) Titanic survival. 3) House price regression. 4) Iris / wine classification. 5) Spam SMS classifier. 6) Customer segmentation (K-Means). 7) Movie dashboard (Streamlit). 8) Data-cleaning pipeline.

Tanglish (Tamil + English) Vilakkam
Beginner projects list: Titanic, house price, spam classifier, K-Means segmentation.

Learn more
YouTube search: beginner machine learning projects with source code

2. Intermediate projects (Weeks 9-24) [Intermediate]

Explanation (English)
1) Churn prediction with XGBoost + SHAP. 2) Credit-card fraud (imbalanced). 3) Sales forecasting. 4) Movie recommender. 5) CNN image classifier with transfer learning. 6) Sentiment analysis with BERT. 7) Resume parser (NER). 8) Deploy model with FastAPI + Docker.

Tanglish (Tamil + English) Vilakkam
Churn, fraud, recommender, CNN, BERT sentiment, deploy.

Learn more
YouTube search: intermediate machine learning projects

3. Advanced projects (Weeks 25-52) [Advanced]

Explanation (English)
1) YOLO custom object detector. 2) Tamil speech-to-text with Whisper fine-tune. 3) RAG chatbot with citations and evals. 4) LoRA-fine-tuned LLM. 5) Multi-agent assistant with tools. 6) Real-time anomaly detection pipeline. 7) Diffusion fine-tune. 8) End-to-end MLOps project with monitoring.

Tanglish (Tamil + English) Vilakkam
YOLO, Tamil ASR, RAG, LoRA, agents, MLOps.

Learn more
YouTube search: advanced ai projects llm rag yolo

4. Kaggle strategy [Intermediate]

Explanation (English)
Start with Getting Started comps; study top notebooks; build a strong CV; feature engineering + GBDT + ensembling; track experiments; learn from post-competition write-ups.

Tanglish (Tamil + English) Vilakkam
Kaggle: CV strong, feature engineering, ensembling.

Learn more
YouTube search: kaggle competition strategy tips
Kaggle Learn: https://www.kaggle.com/learn

5. Interview: Python & coding [Intermediate]

Explanation (English)
Common: list vs tuple, deep vs shallow copy, GIL, generators, decorators, mutable defaults, time complexity, two-sum, sliding window, string manipulation, SQL joins/window functions.

Tanglish (Tamil + English) Vilakkam
Python interview: GIL, decorators, generators, copy.

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YouTube search: python interview questions for data science

6. Interview: statistics & maths [Intermediate]

Explanation (English)
Explain p-value, CLT, bias-variance, MLE, confidence interval, Bayes problem, correlation vs causation, PCA/eigenvectors, gradient descent, regularisation.

Tanglish (Tamil + English) Vilakkam
p-value, CLT, MLE, PCA, gradient descent.

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YouTube search: statistics interview questions data science

7. Interview: machine learning [Intermediate]

Explanation (English)
Bagging vs boosting, overfitting fixes, L1 vs L2, precision vs recall, ROC-AUC, handle imbalance & missing data, feature selection, cross-validation, explain random forest/XGBoost/SVM/KNN, leakage, cold start.

Tanglish (Tamil + English) Vilakkam
ML interview: bagging vs boosting, metrics, imbalance.

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YouTube search: machine learning interview questions and answers

8. Interview: deep learning & LLMs [Advanced]

Explanation (English)
Backprop, vanishing gradients, BatchNorm vs LayerNorm, attention complexity, positional encoding, LoRA, RAG design, hallucination mitigation, tokenisation, KV cache, evaluation of LLM apps.

Tanglish (Tamil + English) Vilakkam
DL/LLM interview: backprop, attention, LoRA, RAG.

Learn more
YouTube search: deep learning llm interview questions

9. Interview: ML system design [Advanced]

Explanation (English)
Clarify goal -> metrics -> data -> features -> model -> training pipeline -> serving (latency, scale) -> monitoring -> iteration. Practice: recommender, fraud detection, search ranking, ad CTR, chatbot.

Tanglish (Tamil + English) Vilakkam
ML system design framework.

Learn more
YouTube search: machine learning system design interview

10. Resume, LinkedIn, GitHub and job search [Beginner]

Explanation (English)
Quantify impact in bullet points, link live demos, keep a strong GitHub README, network, apply to internships, use job boards (LinkedIn, Wellfound, Naukri).

Tanglish (Tamil + English) Vilakkam
Resume-la impact numbers sollunga, GitHub clean-a vaingka.

Learn more
YouTube search: data science resume tips

11. Certifications & degrees (optional) [Beginner]

Explanation (English)
Useful: Google/IBM/DeepLearning.AI certificates, AWS ML Specialty, TensorFlow Developer, Azure AI Engineer. Projects matter more than certificates.

Tanglish (Tamil + English) Vilakkam

Certificates help, aana projects-dhaan mukkiyam.

Learn more
YouTube search: best machine learning certifications